

Chapter 07: Dislocations & Strengthening Mechanisms in Metals

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Materials Science (MIAE 221)

Textbook:

Materials Science and Engineering: An Introduction

(10th Edition) by William D. Callister, Jr. & David G. Rethwisch — Chapter 7

1. Executive Overview & First-Principles Philosophy

In engineering mechanics, pure theoretical physics predicts that metals should possess astronomical strengths: pulling a block of pure iron apart should require a shear stress of approximately $\tau_{\text{theor}} \approx \frac{G}{2\pi} \approx 10,000 \text{ MPa}$ (10 GPa) to simultaneously break all interatomic bonds across a crystalline plane.

Yet in reality, single crystals of pure iron begin to yield plastically at a microscopic shear stress of merely $\approx 1 \text{ MPa}$ —four orders of magnitude ($10,000\times$) lower than theoretical predictions!

This massive discrepancy was solved in 1934 by Taylor, Orowan, and Polanyi: **plastic deformation in metals does not occur by simultaneously shearing entire planes of atoms. It occurs by the sequential motion of line defects called Dislocations (dislocation slip).**

- **The Carpet Analogy:** Imagine moving a heavy 20-foot Persian rug across a floor. If you try to drag the entire rug at once, the friction force from all contacts simultaneously makes it impossible to move. But if you push a small wrinkle (a "dislocation") into one end and kick that wrinkle across the rug, you only overcome friction along one line of carpet at a time. The wrinkle exits the opposite end, and the entire rug has moved forward by one step with almost zero effort!
- **The Grand Axiom of Metallurgical Strengthening:**

$$\text{Strength} \iff \text{Resistance to Dislocation Motion}$$

Because dislocations are the vehicles of plastic deformation, **every metallurgical technique used to strengthen a metal works by placing microstructural obstacles in the path of moving dislocations.**

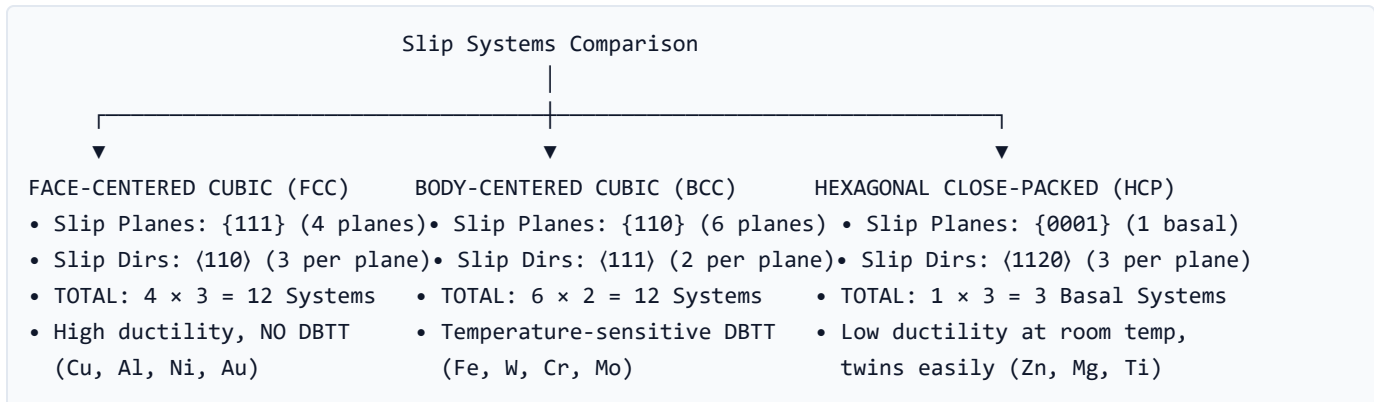
2. Slip Systems in Crystalline Metals (Callister §7.4)

Dislocations do not move randomly through a crystal lattice. They glide along specific crystallographic planes and in specific directions:

- **Slip Plane:** The plane of **highest planar atomic density** (most closely packed plane). The spacing between adjacent close-packed planes (d_{hkl}) is maximized, meaning atomic roughness and the Peierls-Nabarro friction stress required to slide planes over each other are minimized.

- **Slip Direction:** The direction within the slip plane having the **highest linear atomic density** (most closely packed direction). The Burgers vector $\vec{b}$ lies along this direction, minimizing its length $|\vec{b}|$. Because dislocation strain energy scales as $E_{\text{strain}} \propto |\vec{b}|^2$, dislocations always move in directions of shortest lattice translation.
- **Slip System:** The combination of a specific slip plane and an associated slip direction:

$$\text{Slip System} = (\text{Slip Plane}) + [\text{Slip Direction}]$$



1. FCC Metals (12 Slip Systems):

- 4 octahedral 111 close-packed planes, each containing 3 close-packed $\langle 110 \rangle$ directions: $4 \times 3 = 12$ slip systems.
- Because 12 independent slip systems intersect in 3D space, FCC metals (Copper, Aluminum, Gold, Nickel) are **exceptionally ductile and formable** even at cryogenic temperatures. They exhibit **no ductile-to-brittle transition (DBTT)**.

2. BCC Metals (12 Primary Systems + 36 Secondary at High T):

- Slips primarily on 110 planes along $\langle 111 \rangle$ directions ($6 \times 2 = 12$ systems), but also on 112 and 123 planes at elevated temperatures.
- BCC metals lack true close-packed planes ($\text{APF} = 0.68$). The Peierls friction stress is high and strongly temperature-dependent, causing BCC steels to undergo a **Ductile-to-Brittle Transition (DBTT)** at low temperatures!

3. HCP Metals (3 Primary Basal Systems):

- Slips primarily on the single basal plane 0001 in the 3 close-packed $\langle 11\bar{2}0 \rangle$ directions: $1 \times 3 = 3$ slip systems.
- Having fewer than 5 independent slip systems (von Mises criterion for generalized polycrystalline plasticity), room-temperature HCP metals (Zinc, Magnesium) are **brittle under tensile forming** and deform via mechanical **twinning**.

3. Slip in Single Crystals & Schmid's Law (Callister §7.5)

When a single crystal is pulled in uniaxial tension by an axial force F producing tensile stress $\sigma = F/A_0$, the stress resolving along a specific internal slip system is a **shear stress**.

represent the angle between the normal to the slip plane and the applied stress direction, and let λ be the angle between the slip and stress directions, as indicated in Figure 7.7; it can then be shown that for the resolved shear stress τ_R

$$\tau_R = \sigma \cos \phi \cos \lambda \quad (7.2)$$

where σ is the applied stress. In general, $\phi + \lambda \neq 90^\circ$ because it need not be the case that the tensile axis, the slip plane normal, and the slip direction all lie in the same plane.

A metal single crystal has a number of different slip systems that are capable of operating. The resolved shear stress normally differs for each one because the orientation

Figure 7.7 Geometric relationships between the tensile axis, slip plane, and slip direction used in calculating the resolved shear stress for a single crystal.

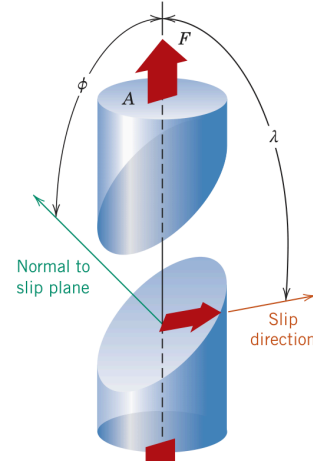


Figure 7.7: Geometrical relationship between tensile loading axis, slip plane normal vector $\vec{n}$ (angle ϕ), and slip direction vector $\vec{s}$ (angle λ) — from Callister & Rethwisch 10th Ed. (Fig. 7.7).

3.1 Schmid's Law Formulation

Let:

- ϕ = angle between the tensile loading axis and the **normal to the slip plane** ($0^\circ \leq \phi \leq 90^\circ$).
- λ = angle between the tensile loading axis and the **slip direction** ($0^\circ \leq \lambda \leq 90^\circ$).

Resolving the axial force along the slip direction ($F_s = F \cos \lambda$) and resolving the cross-sectional area along the inclined slip plane ($A_s = A_0 / \cos \phi$):

$$\tau_R = \frac{F_s}{A_s} = \frac{F \cos \lambda}{A_0 / \cos \phi} = \left(\frac{F}{A_0} \right) \cos \phi \cos \lambda$$

$$\tau_R = \sigma \cos \phi \cos \lambda \quad (\text{Schmid's Law})$$

where:

- τ_R = **Resolved Shear Stress** acting on the slip system.
- $m = \cos \phi \cos \lambda$ = **Schmid Factor** ($0 \leq m \leq 0.50$).
- The maximum possible value of the Schmid factor is $m_{\max} = 0.50$, occurring when $\phi = \lambda = 45^\circ$.

3.2 Critical Resolved Shear Stress (τ_{CRSS}) & Yielding

Plastic deformation initiates when the resolved shear stress on the most favorably oriented slip system reaches a material-specific threshold: the **Critical Resolved Shear Stress** (τ_{CRSS}):

$$\tau_R \geq \tau_{\text{CRSS}}$$

The macroscopic yield strength σ_y of the single crystal is therefore:

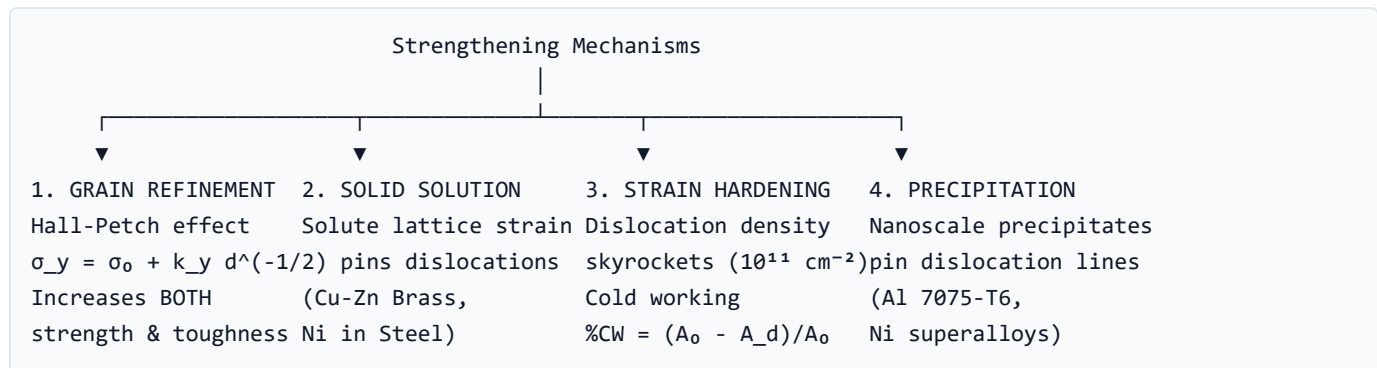
$$\sigma_y = \frac{\tau_{\text{CRSS}}}{(\cos \phi \cos \lambda)_{\text{max}}}$$

- **Orientation Extremes:**

- If the tensile axis is perpendicular to the slip plane ($\phi = 0^\circ \implies \lambda = 90^\circ \implies \tau_R = 0$).
- If the tensile axis is parallel to the slip plane ($\phi = 90^\circ \implies \tau_R = 0$).
- In both extreme orientations, $\tau_R = 0$: **no dislocation slip can occur**, and the single crystal will fracture elastically at very high stresses rather than yielding!

4. The Four Foundational Strengthening Mechanisms (Callister §7.8 – §7.10)

To make a metal stronger, an engineer must impede dislocation motion. Four primary mechanisms are employed in industrial metallurgy:



4.1 Strengthening Mechanism 1: Grain Size Reduction (The Hall-Petch Law)

A polycrystalline metal consists of millions of crystallites (grains) separated by grain boundary interfaces.

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hardness and strength (both yield and tensile) are related to the ease with which plastic deformation can be made to occur, by reducing the mobility of dislocations, the mechanical strength may be enhanced—that is, greater mechanical forces are required to initiate plastic deformation. In contrast, the more unconstrained the dislocation motion, the greater is the facility with which a metal may deform, and the softer and weaker it becomes. Virtually all strengthening techniques rely on this simple principle: *Restricting or hindering dislocation motion renders a material harder and stronger.*

The present discussion is confined to strengthening mechanisms for single-phase metals by grain size reduction, solid-solution alloying, and strain hardening. Deformation and strengthening of multiphase alloys are more complicated, involving concepts beyond the scope of the present discussion; Chapter 10 and Section 11.10 treat techniques that are used to strengthen multiphase alloys.

STRENGTHENING BY GRAIN SIZE REDUCTION

The size of the grains, or average grain diameter, in a polycrystalline metal influences the mechanical properties. Adjacent grains normally have different crystallographic orientations and, of course, a common grain boundary, as indicated in Figure 7.14. During plastic deformation, slip or dislocation motion must take place across this common boundary—say, from grain A to grain B in Figure 7.14. The grain boundary acts as a barrier to dislocation motion for two reasons:

Figure 7.14: Dislocation slip planes terminate at a grain boundary. Leading dislocations pile up against the boundary barrier, producing stress concentrations required to activate slip in the adjoining misoriented grain — from Callister & Rethwisch 10th Ed. (Fig. 7.14).

Physical Barrier Mechanism:

Grain boundaries act as formidable barriers to dislocation slip for two physical reasons:

1. **Crystallographic Misorientation:** When a dislocation traveling across Grain A reaches the boundary, the slip planes in adjacent Grain B are oriented in a different spatial direction. The dislocation cannot cross without abruptly changing direction, requiring high energy.
2. **Atomic Disorder:** The atomic disorder and disrupted bonding in the boundary zone prevent continuous slip plane continuity.

Dislocations pile up against the boundary like a traffic jam. The pile-up generates a concentrated stress field that must reach a critical magnitude to activate new dislocation sources in the neighboring grain. Smaller grains have shorter slip distances, producing smaller pile-ups that require higher applied macroscopic stresses to propagate deformation.

The Hall-Petch Equation:

The yield strength σ_y scales inversely with the square root of the average grain diameter d :

$$\sigma_y = \sigma_0 + k_y d^{-1/2} = \sigma_0 + \frac{k_y}{\sqrt{d}}$$

where:

- d = average grain diameter (in m or mm).

- σ_0 = friction stress (lattice resistance to dislocation motion in a single crystal).
 - k_y = Hall-Petch strengthening coefficient (material constant measuring boundary barrier strength).
 - 🌟 **The Supreme Advantage of Grain Refinement:** Grain size reduction is the **ONLY strengthening mechanism in metallurgy that increases both strength AND fracture toughness/ductility simultaneously!** All other mechanisms (cold working, alloying, precipitation) increase strength at the expense of ductility.
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4.2 Strengthening Mechanism 2: Solid-Solution Strengthening

Introducing substitutional or interstitial solute atoms into a pure solvent metal lattice increases yield strength (e.g., adding Zinc to Copper to create high-strength Brass):

- **Lattice Strain Interaction Mechanism:**
 - An impurity atom of different atomic radius introduces localized elastic strain fields into the host lattice:
 - A **smaller substitutional solute** ($R_{\text{solute}} < R_{\text{solvent}}$) generates a **tensile strain field** around itself.
 - A **larger substitutional solute** ($R_{\text{solute}} > R_{\text{solvent}}$) generates a **compressive strain field** around itself.
 - Recall from Chapter 4 that an edge dislocation possesses a **compressive strain zone above its slip plane** (extra half-plane) and a **tensile strain zone below its slip plane**.
 - Solute atoms diffuse preferentially to positions where their strain fields cancel the dislocation's strain field:
 - Smaller solutes migrate to the compressive zone above the slip plane.
 - Larger solutes migrate to the tensile zone below the slip plane.
 - This mutual strain relief lowers overall system energy, locking the dislocation in place (**Cottrell pinning atmosphere**). An extra applied shear stress is required to pull the dislocation away from its solute cloud!
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4.3 Strengthening Mechanism 3: Strain Hardening (Work Hardening / Cold Work)

Strain hardening is the phenomenon whereby a ductile metal becomes stronger and harder as it is plastically deformed at temperatures below its recrystallization temperature (typically at room temperature).

The Dislocation Multiplication Mechanism:

- During plastic deformation, dislocations do not simply move and escape to surfaces; they **multiply exponentially** through Frank-Read dislocation mills.
- Dislocation density increases from $\rho \approx 10^5 \text{ cm}^{-2}$ in annealed metals to $\rho \approx 10^9 - 10^{11} \text{ cm}^{-2}$ **in heavily cold-worked metals!**
- As dislocation density skyrockets, the average distance between dislocations shrinks. Dislocations intersect, forming jogs and dense, tangled "forest" networks.
- Because dislocations exert repulsive stress fields on one another, moving dislocations become hopelessly pinned and tangled. A progressively higher applied stress is required to force dislocations through the forest,

raising the yield strength σ_y and ultimate tensile strength σ_u while reducing ductility toward zero.

Percent Cold Work () Formulation:

The degree of plastic deformation is quantified by the **Percent Cold Work ()**:

where A_0 is the original cross-sectional area and A_d is the deformed cross-sectional area after cold rolling, forging, or wire drawing.

4.4 Strengthening Mechanism 4: Precipitation Hardening (Age Hardening)

Extensively utilized in high-performance aerospace alloys (e.g., Al 7075-T6, Al 2024, Inconel superalloys):

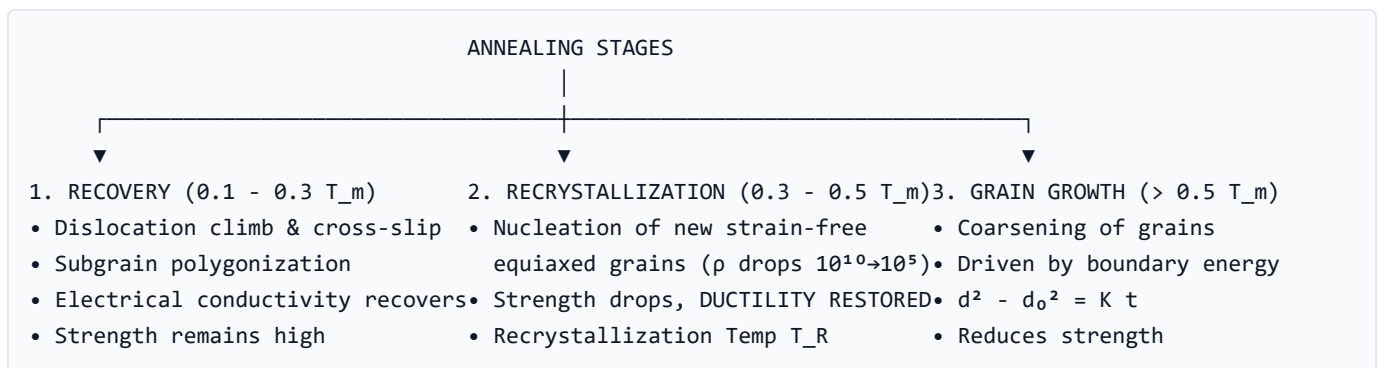
- Extremely fine, uniform, nanoscale second-phase precipitate particles (e.g., Al_2CuMg or $\text{Ni}_3(\text{Al}, \text{Ti})$) are precipitated within the matrix grains through a three-step heat treatment: Solution Treatment $\rightarrow$ Rapid Quenching $\rightarrow$ Controlled Aging.
- Moving dislocations cannot shear through hard precipitates easily; they either shear the precipitate (requiring huge energy) or are forced to bow between precipitates leaving residual dislocation loops (**Orowan Dislocation Bypassing Mechanism**):

$$\tau \propto \frac{Gb}{L}$$

where L is the inter-precipitate spacing. Dense, closely spaced precipitates yield maximum strength!

5. Annealing: Recovery, Recrystallization & Grain Growth (Callister §7.11 – §7.13)

When a heavily cold-worked metal is heated, roughly 5 of the deformation energy stored as elastic strain energy in dislocation networks provides the thermodynamic driving force to reverse the cold-worked state across three sequential stages:



5.1 Stage 1: Recovery ($T \approx 0.1 - 0.3T_m$)

- Dislocation density remains essentially unchanged, but thermal activation permits atoms to diffuse, allowing dislocations to climb vertically and cross-slip.

- Dislocations of opposite sign annihilate each other; remaining dislocations rearrange into low-energy, ordered arrays forming **subgrain polygonized boundaries**.
- Electrical and thermal conductivities recover to pre-cold-worked values; mechanical strength and hardness remain high.

5.2 Stage 2: Recrystallization ($T \approx 0.3 - 0.5T_m$)

- A brand new set of **strain-free, equiaxed grains** nucleates at regions of high strain energy (grain boundaries and deformation bands) and consumes the cold-worked matrix.
- Dislocation density drops by five orders of magnitude ($10^{11} \rightarrow 10^5 \text{ cm}^{-2}$).
- **Mechanical Transformation:** Hardness and tensile strength drop sharply back to annealed baseline values; **tensile ductility () is completely restored!**
- **The Recrystallization Temperature (T_R):** The temperature at which recrystallization reaches completion within exactly **1 hour**:
 - Typically $T_R \approx 0.3 - 0.4T_m$ for pure metals; $T_R \approx 0.5 - 0.7T_m$ for commercial alloys.
 - Factors lowering T_R : higher prior (more stored strain energy), higher metal purity (solutes pin migrating boundaries), longer heating time.

5.3 Stage 3: Grain Growth ($T > 0.5T_m$)

- After recrystallization completes, if heating continues, the newly formed grains coarsen.
- **Driving Force:** Reduction in total grain boundary surface area and energy. Smaller grains with concave boundary curvature shrink and disappear while larger grains grow.
- The average grain diameter d scales with annealing time t according to:

$$d^n - d_0^n = Kt$$

where d_0 is initial grain diameter, K is a temperature-dependent Arrhenius rate constant, and $n \approx 2$ for pure metals.

- Because larger grains lower yield strength (via Hall-Petch), **excessive grain growth is avoided in manufacturing!**

6. Comprehensive Step-by-Step Problem Walkthroughs

6.1 Problem 1: Schmid's Law & Single Crystal Yielding

Problem Statement: A single crystal of an FCC metal is oriented such that a uniaxial tensile stress $\sigma = 4.50 \text{ MPa}$ is applied along the $[001]$ crystallographic direction.

1. Determine the resolved shear stress τ_R acting on the (111) slip plane in the $[10\bar{1}]$ slip direction.
2. If the critical resolved shear stress for this metal is $\tau_{\text{CRSS}} = 2.00 \text{ MPa}$, determine whether the single crystal will yield under this load.
3. Calculate the macroscopic yield strength σ_y of the crystal in this orientation.

Step 1: Compute Angles ϕ and λ Using Vector Dot Products

Recall the angle formula between two vectors $\vec{u} = [u_1, u_2, u_3]$ and $\vec{v} = [v_1, v_2, v_3]$:

$$\cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}||\vec{v}|} = \frac{u_1v_1 + u_2v_2 + u_3v_3}{\sqrt{u_1^2 + u_2^2 + u_3^2}\sqrt{v_1^2 + v_2^2 + v_3^2}}$$

- **Angle ϕ (Between Tensile Axis $[001]$ and Slip Plane Normal $[111]$):**

$$\cos \phi = \frac{(0)(1) + (0)(1) + (1)(1)}{\sqrt{0^2 + 0^2 + 1^2}\sqrt{1^2 + 1^2 + 1^2}} = \frac{1}{(1)\sqrt{3}} = \frac{1}{\sqrt{3}} \approx 0.5774$$

$$\phi = \arccos(0.5774) = 54.74^\circ$$

- **Angle λ (Between Tensile Axis $[001]$ and Slip Direction $[10\bar{1}]$):**

$$\cos \lambda = \frac{(0)(1) + (0)(0) + (1)(-1)}{\sqrt{0^2 + 0^2 + 1^2}\sqrt{1^2 + 0^2 + (-1)^2}} = \frac{-1}{(1)\sqrt{2}} = -\frac{1}{\sqrt{2}} \approx -0.7071$$

Taking the magnitude for resolved shear stress ($|\cos \lambda| = \frac{1}{\sqrt{2}} \approx 0.7071$):

$$\lambda = 45.00^\circ$$

Step 2: Compute the Resolved Shear Stress τ_R

$$\tau_R = \sigma \cos \phi |\cos \lambda| = (4.50 \text{ MPa}) \left(\frac{1}{\sqrt{3}} \right) \left(\frac{1}{\sqrt{2}} \right) = \frac{4.50}{\sqrt{6}} = \frac{4.50}{2.4495} = 1.837 \text{ MPa}$$

Step 3: Yield Condition Check

Compare τ_R to τ_{CRSS} :

$$\tau_R = 1.84 \text{ MPa} < \tau_{\text{CRSS}} = 2.00 \text{ MPa}$$

Conclusion

: The single crystal **will NOT yield** under $\sigma = 4.50 \text{ MPa}$; it deforms purely elastically.

Step 4: Calculate Macroscopic Yield Strength σ_y

$$\sigma_y = \frac{\tau_{\text{CRSS}}}{\cos \phi \cos \lambda} = \frac{2.00 \text{ MPa}}{\frac{1}{\sqrt{6}}} = 2.00 \times \sqrt{6} = 2.00 \times 2.4495 = 4.90 \text{ MPa}$$

Yielding will initiate when the applied tensile stress reaches $\sigma = 4.90 \text{ MPa}$.

6.2 Problem 2: Hall-Petch Grain Size Strengthening in Steel

Problem Statement: The yield strength of a structural low-carbon steel was measured for two different average grain diameters:

- For average grain diameter $d_1 = 0.050 \text{ mm}$ ($50 \mu\text{m}$), yield strength $\sigma_{y1} = 160 \text{ MPa}$.
 - For average grain diameter $d_2 = 0.010 \text{ mm}$ ($10 \mu\text{m}$), yield strength $\sigma_{y2} = 260 \text{ MPa}$.
1. Determine the constants σ_0 and k_y in the Hall-Petch equation.
 2. Predict the yield strength of the steel if the grain size is refined to $d_3 = 0.004 \text{ mm}$ ($4 \mu\text{m}$).

Step 1: Formulate the Hall-Petch Linear System

$$\sigma_y = \sigma_0 + k_y d^{-1/2}$$

Calculate $d^{-1/2}$ for both states:

- $d_1^{-1/2} = (0.050 \text{ mm})^{-1/2} = \frac{1}{\sqrt{0.050}} = 4.472 \text{ mm}^{-1/2}$
- $d_2^{-1/2} = (0.010 \text{ mm})^{-1/2} = \frac{1}{\sqrt{0.010}} = 10.000 \text{ mm}^{-1/2}$

Set up the two simultaneous algebraic equations:

$$\sigma_0 + 4.472k_y = 160 \quad \text{--- (Eq. 1)} \quad \sigma_0 + 10.000k_y = 260 \quad \text{--- (Eq. 2)}$$

Step 2: Solve for k_y and σ_0

Subtract (Eq. 1) from (Eq. 2):

$$(10.000 - 4.472)k_y = 260 - 160$$

$$5.528k_y = 100 \implies k_y = \frac{100}{5.528} = 18.09 \text{ MPa} \cdot \text{mm}^{1/2}$$

Substitute k_y into (Eq. 1):

$$\sigma_0 + 4.472(18.09) = 160 \implies \sigma_0 + 80.90 = 160 \implies \sigma_0 = 160 - 80.90 = 79.1 \text{ MPa}$$

Step 3: Predict Yield Strength for $d_3 = 0.004 \text{ mm}$ ($4 \mu\text{m}$)

$$d_3^{-1/2} = \frac{1}{\sqrt{0.004}} = 15.811 \text{ mm}^{-1/2}$$

$$\sigma_{y3} = \sigma_0 + k_y d_3^{-1/2} = 79.1 \text{ MPa} + (18.09 \text{ MPa} \cdot \text{mm}^{1/2}) \times (15.811 \text{ mm}^{-1/2})$$

$$\sigma_{y3} = 79.1 + 286.0 = 365.1 \text{ MPa} \approx 365 \text{ MPa}$$

Engineering Impact

: Refining grains from $50 \mu\text{m}$ down to $4 \mu\text{m}$ more than **doubles the yield strength** from 160 MPa to 365 MPa with zero loss of ductility!

7. Critical Concordia Exam Traps & Red Flags

-  **Trap 1: Confusion Between ϕ and λ in Schmid's Law:**

- ϕ is the angle between the load axis and the **normal to the slip plane**.
- λ is the angle between the load axis and the **slip direction vector**.
- In 3D space, $\phi + \lambda \neq 90^\circ$! (They only sum to 90° in 2D). Do not assume $\cos \lambda = \sin \phi$. You must compute both dot products independently!
- **⚠ Trap 2: Units of Grain Diameter in Hall-Petch:** The coefficient k_y has units of $\text{MPa} \cdot \text{mm}^{1/2}$ or $\text{MPa} \cdot \text{m}^{1/2}$. If you compute k_y using grain diameter in millimeters (mm), you **must input subsequent grain diameters in millimeters**! Converting d to meters midway will produce answers off by a factor of $\sqrt{1000} \approx 31.6$.
- **⚠ Trap 3: Ductility vs. Strength Trade-Off:** Remember: Solid solution strengthening, cold work, and precipitation hardening all increase yield strength while **reducing ductility**. Grain size reduction is the **only** strengthening mechanism that increases both strength and ductility.
- **⚠ Trap 4: Cold Work Area vs. Diameter Formula:** . Students frequently write $\frac{d_0 - d_d}{d_0} \times 100$ (forgetting to square diameters). is based on **cross-sectional area**, NOT diameter!